IGCSE ECONOMICS · SECTIONS 2.1 TO 2.5

Study Companion: Markets, Demand, Supply and Price Changes

Learn the key words, practise multiple choice, then build full Paper 2 answers step by step. Model answers and student mark schemes are at the back, so you can mark your own work.

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Paper 2 toolkit

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Paper 2 toolkit

Every Paper 2 question in this guide has four parts, like a Section B question: **(a) 2 marks, (b) 4 marks, (c) 6 marks, (d) 8 marks = 20 marks**. The command word tells you what kind of answer to write.

Command word	What it means	How to build your answer	Time
Define / State what is meant by [2]	Give the exact meaning of a term.	One sentence with the **two key ideas** of the definition. No example needed.	3 min
Explain [4]	Say **why** or **how** something happens.	Two points. For each: **name it** (1 mark), then **say why it causes the effect** (1 mark).	6 min
Analyse [6]	Show the steps from cause to effect.	A **chain of reasoning**: each link is one step. Use "so", "this means", "as a result". Add a labelled diagram.	9 min
Discuss [8]	Give both sides of an argument.	Two developed points for "**might**" and two for "**might not**". End with a short conclusion: "It depends on...".	12 min

How to write a chain of reasoning

Examiners give marks for each **link** in the chain. Do not jump from the start to the end.

Cause → **What happens to the curve** → **Shortage or surplus at the old price** → **Price changes** → **Extension / contraction** → **New P and Q**

Chuỗi lập luận: nguyên nhân → đường cầu/cung dịch chuyển → thiếu hụt hoặc dư thừa → giá thay đổi → di chuyển dọc đường → giá và lượng cân bằng mới.



Blank axes: start every diagram like this.

Diagram checklist

- ☐ Axes labelled **Price (₫)** and **Quantity**
- ☐ Demand in **blue** (slopes down), supply in **red** (slopes up)
- ☐ Every curve labelled: D1, D2, S1, S2
- ☐ An arrow to show the shift
- ☐ Dotted lines to the axes: P1, Q1, then P2, Q2
- ☐ Shortage or surplus marked at the old price

How to mark your own work

1. Finish the whole answer first, then use a different colour pen.
2. Use the mark scheme table at the back. Tick each row only if the idea is **clearly** there. Different words are fine if the meaning is the same.
3. Write the mark next to each tick, then the total. Do not go over the maximum.
4. Compare with the model answer. Write one thing you did well and one step that is missing, then rewrite that step.

2.1 SECTION 2.1
The role of markets in allocating resources

Key words

Market
Any place or system where buyers and sellers come together to exchange goods and services.

Simply: where people buy and sell. It can be online (Shopee) or a real place (Bến Thành Market).

thị trường

Product market
A market where households buy goods and services from firms.

Simply: where you buy things to use, like phones, food and bubble tea.

thị trường hàng hóa và dịch vụ

Factor market
A market where firms buy or hire factors of production (land, labour, capital, enterprise) from their owners.

Simply: where firms get what they need to make things: workers, land, machines.

thị trường yếu tố sản xuất

Buyers and sellers
Buyers (consumers) create demand. Sellers (producers) create supply. Price is agreed between them.

Simply: buyers want the product and can pay; sellers make it to earn profit.

người mua · người bán

Resource allocation
Deciding how scarce resources are used: what to produce, how to produce it, and for whom.

Simply: we cannot make everything, so we choose.

phân bổ nguồn lực

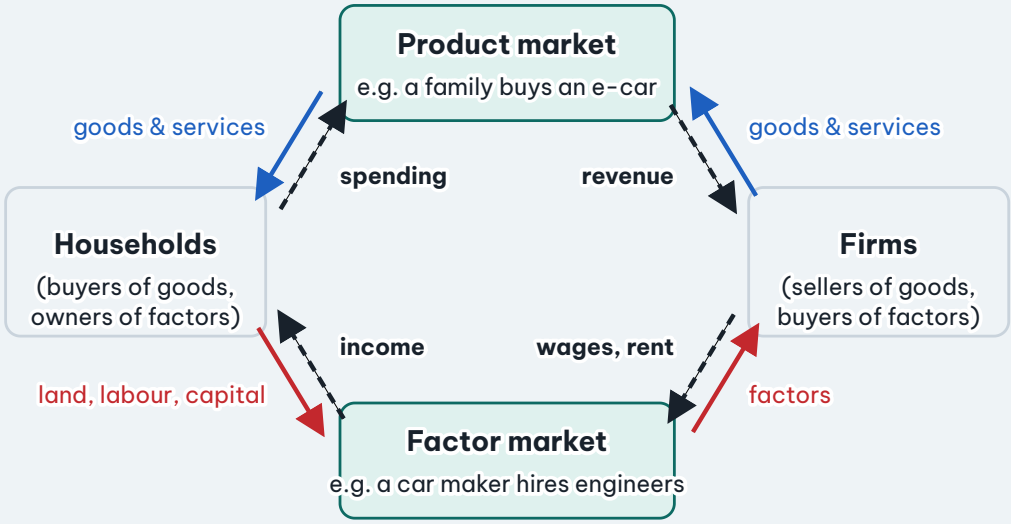
Price mechanism
The way changes in price, set by demand and supply, decide how resources are allocated in a market economy.

Simply: prices tell firms what to make and decide who gets the product.

cơ chế giá

Memory hook: product market or factor market?

Product market = **output**, bought by **households** (Bến Thành Market, Shopee, Phúc Long). **Factor market** = **inputs**, bought by **firms** (a factory in the Saigon Hi-Tech Park hiring engineers, a shop renting land in District 1).



Solid arrows: goods, services and factors. Dashed arrows: money, which always flows the other way.

Memory hook: S.I.R. (how the price mechanism works)

S	**Signal.** A higher price tells producers that consumers want more.
I	**Incentive.** A higher price means more profit, so firms move resources into making more.
R	**Rationing.** A higher price reduces the quantity demanded, so the product goes to people most willing and able to pay.

Tín hiệu (signal) · Khuyến khích (incentive) · Phân phối/hạn chế (rationing)

Paper 1 practice: multiple choice

1. What is the main role of a market?

A to let the government decide all prices

B to bring buyers and sellers together to trade

C to give every worker the same income

D to remove the problem of scarcity

2. Which exchange takes place in a factor market?

A a family buys a new motorbike

B a student buys bubble tea at a café

C a tourist pays for a hotel room in Hà Nội

D a garment factory pays rent for a building

3. What is the role of buyers in a market?

A They show demand by being willing and able to pay.

B They set the price that every firm must charge.

C They make goods and services to earn a profit.

D They decide the lowest wage that firms may pay.

4. Consumers start to like a product more. How does the price mechanism signal this to producers?

A The price of the product falls.

B The government pays firms a subsidy.

C The price of the product rises.

D The supply of the product contracts.

Paper 2 practice: structured question

CONTEXT

Electric cars in Việt Nam

More families in Việt Nam want electric cars. They are quiet, cheap to run and fashionable. Families buy cars from car makers in the **product market**. To make more cars, a car maker must hire engineers and factory workers, rent land and buy machines in **factor markets**.

Last year some popular models had a waiting list of more than three months.

One car maker (practice figures)	Cars ordered per month	Cars produced per month
January	5,000	5,000
June	8,000	5,500

Define **(a) Define the term *market*.** **[2]**

To get 2 marks you need two ideas: **who** is in a market, and **what they do**.

1. **Who?** *A market is where* _____ *and* _____ *come together…*
2. **What do they do?** *…to* _____ *goods and services.*

Word bank: buyers | sellers | exchange | trade | goods and services

> **Watch out:** "a shop" or "a place to buy things" is not enough. A market does not need to be a real place.

Explain **(b) Explain the difference between a product market and a factor market. Use an example of each from the context.** **[4]**

To get 4 marks: explain the product market (1) + example (1); explain the factor market (1) + example (1).

1. **Product market:** *In a product market, households buy* _____ *from* _____.
2. **Example from the context:** *For example,* _____.
3. **Factor market:** *In contrast, in a factor market, firms buy or hire* _____ *such as* _____.
4. **Example from the context:** *For example, the car maker* _____.

Word bank: households | firms | goods and services | factors of production | land | labour | capital | in contrast

> **Watch out:** "in contrast" shows the examiner you are comparing. Use examples from the context, not your own, when the question says "from the context".

Analyse **(c) Analyse how the price mechanism allocates scarce resources when consumer demand for electric cars increases.** **[6]**

To get 6 marks: six links in a chain. Use **S.I.R.**: signal, incentive, rationing. Draw a diagram.

1. **Shift:** *An increase in demand shifts the demand curve to the _____, from D1 to D2.*
2. **Old price:** *At the old price P1, quantity demanded is greater than quantity supplied, so there is a _____.*
3. **Signal:** *The price rises. This signals to car makers that _____.*
4. **Incentive:** *Higher prices mean more profit, so firms move more _____ (workers, machines, land) into making electric cars. Supply _____.*
5. **Rationing:** *The higher price makes demand _____, so the cars go to buyers who are most willing and able to pay.*
6. **Result:** *A new equilibrium is reached at a higher price P2 and higher quantity Q2. More resources are now used for _____.*



Draw: S, D1, D2, P1, Q1, P2, Q2 and the shortage at P1.

Word bank: shortage | signal | incentive | rationing | extends | contracts | factors of production | profit

Watch out: after the shift, supply **extends** (a movement along S). The supply curve does **not** shift.

lượng cung tăng (di chuyển lên dọc đường cung) = supply extends

Discuss **(d) Discuss whether a higher price for electric cars will always lead to more resources being used to make them.** **[8]**

To get 8 marks: up to 4 marks for why it **might** and up to 4 marks for why it **might not**. Each point = make the point (1) + explain it (1).

Why it might

A higher price might lead to more resources because...

- profit and incentive
- new firms enter

Why it might not

However, more resources might not be used because...

- time to build factories
- shortage of engineers or batteries
- demand may not last

1. **Might (point 1):** *A higher price increases the profit on each car, so* _____ .
2. **Might (point 2):** *The higher price also signals to other firms that* _____, *so they may enter the market.*
3. **Might not (point 1):** *However, it takes time to* _____, *so in the short run...*
4. **Might not (point 2):** *Also, some resources are scarce, for example* _____ , *so...*
5. **Conclusion:** *Overall, a higher price is likely to lead to more resources in the long run, but it depends on* _____.

Watch out: the word **"always"** is a clue. You must show cases where it does **not** happen.

2.2 SECTION 2.2 Demand

Key words

Demand

The quantity of a good or service that consumers are willing and able to buy at each price in a given time period.

Simply: how much people want to buy **and** can pay for, at each price.

cầu

Effective demand

The wish to buy a product, backed by the ability to pay for it.

Simply: wanting it **and** having the money. Wanting alone is not demand.

cầu có khả năng thanh toán

Individual and market demand

Individual demand is the demand of one consumer. Market demand is the total demand of all consumers in the market.

Simply: add up everyone's demand at each price to get market demand.

cầu cá nhân · cầu thị trường

Law of demand

When price falls, quantity demanded rises; when price rises, quantity demanded falls (other things staying the same).

Simply: cheaper = people buy more. So the demand curve slopes down.

luật cầu

Extension of demand

A rise in quantity demanded caused by a fall in the product's own price: a movement down along the demand curve.

Simply: price down → buy more → move along the curve.

lượng cầu tăng (di chuyển xuống dọc đường cầu)

Contraction of demand

A fall in quantity demanded caused by a rise in the product's own price: a movement up along the demand curve.

Simply: price up → buy less → move along the curve.

lượng cầu giảm (di chuyển lên dọc đường cầu)

Increase / decrease in demand

A shift of the whole demand curve right (increase) or left (decrease), caused by a change in a condition of demand, not the product's own price.

Simply: people want more (or less) at **every** price.

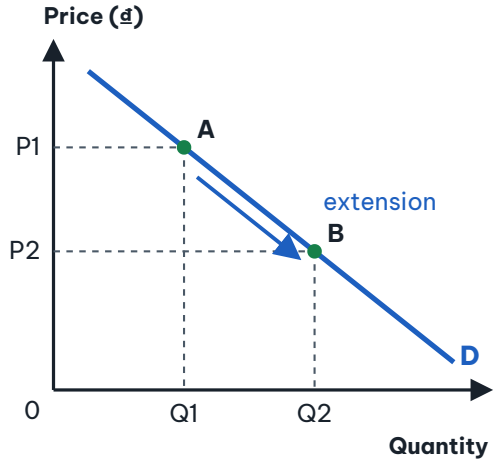
cầu tăng (đường cầu dịch chuyển sang phải) · cầu giảm (đường cầu dịch chuyển sang trái)

Substitutes and complements

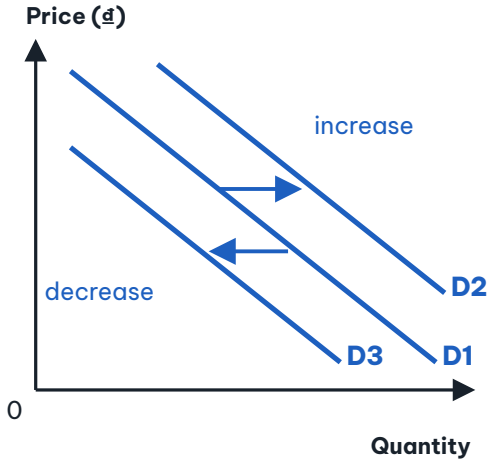
Substitutes are used instead of each other (coffee and tea). Complements are used together (petrol and petrol motorbikes).

Simply: substitute = "or"; complement = "and".

hàng hóa thay thế · hàng hóa bổ sung



Movement along the curve: only the product's own price changes.



Shift of the curve: a condition of demand changes (PASIFIC).

Memory hook: PASIFIC (conditions of demand that shift the curve)

Yếu tố quyết định cầu (conditions of demand)

P	**Population.** More consumers → demand shifts right.
A	**Advertising.** A successful campaign → demand shifts right.
S	**Substitutes.** The price of a substitute rises → demand for our product shifts right.
I	**Income.** Higher income → demand for **normal goods** shifts right; demand for **inferior goods** shifts left.
F	**Fashion and tastes.** The product becomes popular → demand shifts right.
I	**Interest rates.** Lower rates → cheaper loans → demand for things bought on credit (cars, houses) shifts right.
C	**Complements.** The price of a complement falls → demand for our product shifts right.

Paper 1 practice: multiple choice

5. What causes a movement along the demand curve for laptops?

A a rise in consumers' incomes

B an advertising campaign for laptops

C a rise in the price of tablets

D a fall in the price of laptops

6. Coffee and tea are substitutes. The price of tea rises. What happens in the coffee market?

A The demand curve for coffee shifts right.

B The demand curve for coffee shifts left.

C Demand for coffee contracts along the curve.

D The supply curve for coffee shifts right.

7. Which change is an extension of demand?

A Price rises from 30,000₫ to 45,000₫; quantity demanded falls from 100 to 80.

B Incomes rise; people buy 30% more at the same price of 20,000₫.

C Price falls from 40,000₫ to 25,000₫; quantity demanded rises from 50 to 85.

D Tastes change; people buy 30% less at the same price of 20,000₫.

8. What is effective demand?

A wishing to own a product, without the money

B wanting a product and having money to buy it

C the total amount of a product that firms make

D the amount the government tells people to buy

Paper 2 practice: structured question

CONTEXT

Motorbikes in Hồ Chí Minh City

Most people in Hồ Chí Minh City travel by motorbike. There are two main types: **petrol motorbikes** and **electric motorbikes**. Electric motorbikes are cheaper to run, and young people see them as modern and clean. Incomes in Việt Nam are rising. Some cities are planning to limit petrol motorbikes in the city centre.

Price of petrol (practice figures)	₫ per litre
January	24,000
June	20,000

Define **(a) Define *effective demand*.** **[2]**

To get 2 marks you need two ideas: the **wish** to buy, and the **ability** to pay.

1. **Wish:** *Effective demand is the* _____ *to buy a product…*
2. **Ability:** *…backed by* _____.

Word bank: willing · able · backed by · money / income

> **Watch out:** "wanting a product" alone gets only 1 mark. A student who wants a car but has no money has no effective demand.

Explain **(b) Explain two non-price factors that could cause an increase in demand for electric motorbikes in Việt Nam.** **[4]**

To get 4 marks: factor 1 named (1) + why it shifts demand right (1); factor 2 named (1) + why (1). Use PASIFIC and the context.

1. **Factor 1:** *One factor is* _____.
2. **Why:** *This means that at every price, consumers* _____, *so the demand curve shifts to the right.*
3. **Factor 2:** *A second factor is* _____.
4. **Why:** *Because* _____, *more people buy electric motorbikes at each price, so demand increases.*

Word bank: rising incomes · normal good · price of petrol rises · complement · substitute · tastes / fashion · advertising · at every price

> **Watch out:** a fall in the price of electric motorbikes is **not** a non-price factor. It causes an extension of demand, not an increase.

Analyse

(c) Analyse how a decrease in the price of petrol affects the market for petrol motorbikes and the market for electric motorbikes. [6]

To get 6 marks: three links for each market. Name the relationships first: petrol is a **complement** to petrol motorbikes; petrol motorbikes and electric motorbikes are **substitutes**. Draw two diagrams.

1. **Relationships:** *Petrol and petrol motorbikes are _____. Petrol motorbikes and electric motorbikes are _____.*
2. **Cost of running:** *Cheaper petrol makes a petrol motorbike _____ to run.*
3. **Petrol motorbike market:** *Demand for petrol motorbikes increases, so the demand curve shifts _____.*
4. **Petrol motorbike result:** *Equilibrium price and quantity of petrol motorbikes both _____.*
5. **Electric motorbike market:** *Some consumers switch away from electric motorbikes, so demand _____ and the curve shifts _____.*
6. **Electric motorbike result:** *Equilibrium price and quantity of electric motorbikes both _____.*



Diagram 1: petrol motorbikes.



Diagram 2: electric motorbikes.

Watch out: the change is in the **petrol** market, but the question asks about the **motorbike** markets. In both motorbike markets it is **demand** that shifts. Supply does not change.

Discuss **(d) Discuss whether rising incomes in Việt Nam will lead to an increase in demand for motorbikes.** **[8]**

To get 8 marks: two developed points for "will" and two for "will not", then a conclusion.

Why it might

- motorbikes are a normal good for most people
- more people can afford their own bike, or a better one

Why it might not

- richer families may switch to cars
- other factors: metro, city limits on petrol bikes, traffic

1. **Might (point 1):** *For most families a motorbike is a normal good, so when incomes rise _____.*
2. **Might (point 2):** *Higher incomes also mean that _____ can afford their own motorbike.*
3. **Might not (point 1):** *However, as incomes rise further, some families may buy a car instead, because _____. For them the motorbike is an _____ good.*
4. **Might not (point 2):** *Also, other conditions of demand may change at the same time, such as _____.*
5. **Conclusion:** *Demand will probably rise for _____ but may fall for _____, so the overall effect depends on…*

hàng hóa thông thường = normal good · hàng hóa thứ cấp = inferior good

2.3 SECTION 2.3 Supply

Key words

Supply

The quantity of a good or service that producers are willing and able to sell at each price in a given time period.

Simply: how much firms want to sell **and** can make, at each price.

cung

Individual and market supply

Individual supply is the supply of one firm. Market supply is the total supply of all firms in the market.

Simply: add up every firm's supply at each price.

cung cá nhân · cung thị trường

Law of supply

When price rises, quantity supplied rises; when price falls, quantity supplied falls (other things staying the same).

Simply: higher price = more profit = firms sell more. So the supply curve slopes up.

luật cung

Extension of supply

A rise in quantity supplied caused by a rise in the product's own price: a movement up along the supply curve.

Simply: price up → sell more → move along the curve.

lượng cung tăng (di chuyển lên dọc đường cung)

Contraction of supply

A fall in quantity supplied caused by a fall in the product's own price: a movement down along the supply curve.

Simply: price down → sell less → move along the curve.

lượng cung giảm (di chuyển xuống dọc đường cung)

Increase / decrease in supply

A shift of the whole supply curve right (increase) or left (decrease), caused by a change in a condition of supply, not the product's own price.

Simply: firms sell more (or less) at **every** price.

cung tăng (đường cung dịch chuyển sang phải) · cung giảm (đường cung dịch chuyển sang trái)

Indirect tax

A tax on spending, paid to the government by the seller, such as VAT or a tax on soft drinks.

Simply: it adds to the firm's costs, so supply decreases.

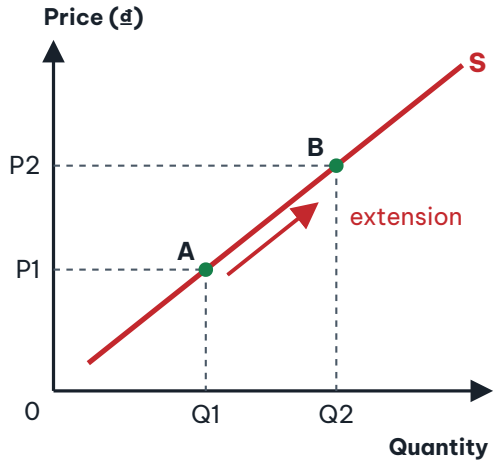
thuế gián thu

Subsidy

A payment from the government to producers that lowers their costs of production.

Simply: government money for firms, so supply increases.

trợ cấp



Movement along the curve: only the product's own price changes.



Shift of the curve: a condition of supply changes (PINTS WC).

Memory hook: PINTS WC (conditions of supply that shift the curve)

Yếu tố quyết định cung (conditions of supply)

P	**Productivity.** Workers produce more per hour → lower cost per unit → supply shifts right.
I	**Indirect taxes.** A higher tax → higher costs → supply shifts left.
N	**Number of firms.** More firms enter the market → supply shifts right.
T	**Technology.** Better machines or robots → lower cost per unit → supply shifts right.
S	**Subsidies.** Government payment to firms → lower costs → supply shifts right.
W	**Weather** and natural disasters. Good weather → bigger harvest → supply shifts right. Floods or drought → supply shifts left.
C	**Costs of production.** Higher wages or raw material prices → less profit per unit → supply shifts left.

Paper 1 practice: multiple choice

9. Which event shifts the supply curve of rice to the right?

A Farmers start using new harvesting machines.

B A long drought damages the rice fields.

C Wages paid to farm workers increase.

D The government adds a tax on fertiliser.

10. The government gives a subsidy to solar panel makers. What happens?

A Demand for solar panels shifts to the left.

B Supply contracts along the supply curve.

C Producers' costs of production increase.

D The supply curve shifts to the right.

11. What causes an extension of supply of smartphones?

A a fall in the cost of microchips

B new robots in assembly factories

C a rise in the price of smartphones

D a cut in the tax on smartphone firms

12. The government raises the indirect tax on soft drinks. What is the effect on soft drink producers?

A They supply more at every price.

B The supply curve shifts to the left.

C The demand curve shifts to the right.

D Supply extends along the supply curve.

Paper 2 practice: structured question

CONTEXT

Coffee in the Central Highlands and laptops in Bắc Ninh

Coffee. Việt Nam is one of the world's biggest coffee producers. Most coffee is grown on small farms in Đắk Lắk and other Central Highlands provinces. In some years, drought damages the coffee trees. Fertiliser and wages have become more expensive, and some farmers are cutting down coffee trees to grow durian instead.

Laptops. Factories in Bắc Ninh assemble laptops. Some factories are now using robots to do work that people did before.

Coffee farm costs (practice figures)	Last year	This year
Fertiliser, ₫ per 50 kg bag	600,000	750,000
Farm worker wage, ₫ per day	250,000	300,000

Define **(a) Define the term *supply*.** **[2]**

To get 2 marks you need: **willing and able** to sell (1) + **at each price, in a time period** (1).

1. **Who and what:** *Supply is the quantity of a product that producers are _____ to sell…*
2. **When:** *…at _____ in a given time period.*

> **Watch out:** supply is about **sellers**, not buyers. "The amount of goods in a shop" is not a definition of supply.

Explain **(b) Explain two factors that could cause a decrease in the supply of coffee beans.** **[4]**

To get 4 marks: factor 1 (1) + why it shifts supply left (1); factor 2 (1) + why (1). The context gives you three good factors. Use PINTS WC.

1. **Factor 1:** *One factor is _____, for example _____.*
2. **Why:** *This means farmers can grow less coffee at every price, so the supply curve shifts to the _____.*
3. **Factor 2:** *A second factor is a rise in costs of production, such as _____.*
4. **Why:** *Each kilogram costs more to produce, so it is _____ profitable and farmers supply less at each price.*



Optional: show S1 shifting left to S2.

Word bank: drought | costs of production | fertiliser | wages | fewer farmers | less profitable | at every price | shifts left

Watch out: a fall in the **price of coffee** causes a contraction of supply, not a decrease. It is not a correct answer here.

Analyse **(c) Analyse how the introduction of robots in factories affects the supply and the price of laptops.** **[6]**

To get 6 marks: six links from robots to the new price. Draw a diagram with a surplus at the old price.

1. **Cause:** *Robots are new technology, so each worker can produce _____ (productivity rises).*
2. **Costs:** *This lowers the _____ of each laptop.*
3. **Shift:** *Firms are willing and able to supply more at every price, so supply shifts to the _____, from S1 to S2.*
4. **Old price:** *At the old price P1, quantity supplied is greater than quantity demanded, so there is a _____.*
5. **Price change:** *Firms cut prices to sell their stock. As the price falls, demand _____.*
6. **Result:** *The new equilibrium has a _____ price (P2) and a _____ quantity (Q2).*



Draw: D, S1, S2, P1, Q1, P2, Q2 and the surplus at P1.

Word bank: technology · productivity · cost per unit · surplus · extends · lower price · higher quantity

Watch out: the question asks about **supply and price**. Do not stop at "supply shifts right". Finish the chain to the new price.

Discuss **(d) Discuss whether a government subsidy to coffee farmers will increase the supply of coffee.** **[8]**

To get 8 marks: two developed points for "will" and two for "will not", then a conclusion.

Why it might

- lower costs → more profit → supply shifts right
- money for better plants, fertiliser, water pumps

Why it might not

- coffee trees take 3 to 4 years to grow
- drought or switching to durian may cancel it out
- subsidy too small, or not used for coffee

1. **Might (point 1):** *A subsidy lowers farmers' _____, so coffee is more profitable and _____.*
2. **Might (point 2):** *Farmers could use the money to _____, which raises productivity.*
3. **Might not (point 1):** *However, new coffee trees take _____, so in the short run…*
4. **Might not (point 2):** *Also, other factors such as _____ could reduce supply more than the subsidy increases it.*
5. **Conclusion:** *A subsidy will probably increase supply in the long run if _____, but…*

2.4 SECTION 2.4 Price determination

Key words

Market equilibrium

The situation where quantity demanded equals quantity supplied, so there is no shortage or surplus.

Simply: buyers want to buy exactly what sellers want to sell.

cân bằng thị trường

Equilibrium price

The price at which quantity demanded equals quantity supplied. The market "clears": there is no pressure for the price to change.

Simply: the price where the demand and supply curves cross.

giá cân bằng

Disequilibrium

Any price where quantity demanded and quantity supplied are not equal, so there is a shortage or a surplus.

Simply: the price is "wrong", so it will change.

mất cân bằng thị trường

Shortage (excess demand)

When the price is **below** equilibrium, quantity demanded is greater than quantity supplied. Buyers compete, so the price is pushed up.

Simply: not enough for everyone → price rises.

thiếu hụt (dư cầu)

Surplus (excess supply)

When the price is **above** equilibrium, quantity supplied is greater than quantity demanded. Sellers have unsold stock, so they cut the price.

Simply: too much left over → price falls.

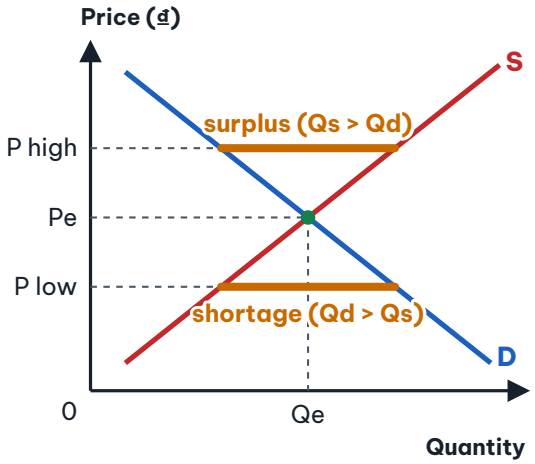
dư thừa (dư cung)

Price mechanism (S.I.R.)

Price changes act as a **signal** to producers and consumers, give an **incentive** to change how much they supply, and **ration** scarce products.

Simply: price moves until the shortage or surplus is gone.

cơ chế giá: tín hiệu · khuyến khích · phân phối



Above Pe: surplus, price falls. Below Pe: shortage, price rises.

How to read a shortage or surplus

1. Find the price on the price axis.
2. Go across to the demand curve and read down: that is **Qd**.
3. Go across to the supply curve and read down: that is **Qs**.
4. Subtract: shortage = Qd − Qs; surplus = Qs − Qd.

Paper 1 practice: multiple choice

13. When is a market in equilibrium?

A when the government fixes the price

B when firms' costs equal their revenue

C when quantity demanded equals quantity supplied

D when all consumers can afford the product

14. The price of a product is above its equilibrium price. What will happen?

A There is a surplus and the price will fall.

B There is a shortage and the price will rise.

C There is a surplus and the price will rise.

D There is a shortage and the price will fall.

15. There is a shortage of concert tickets. How does a free market remove it?

A The price falls; demand extends and supply contracts.

B The supply curve shifts left until it meets demand.

C The demand curve shifts right until it meets supply.

D The price rises; demand contracts and supply extends.

16. Incomes rise, and the product is a normal good. What happens to the equilibrium price and quantity?

A Price falls and quantity falls.

B Price rises and quantity rises.

C Price rises and quantity falls.

D Price falls and quantity rises.

Paper 2 practice: structured question

CONTEXT

The durian market in the Mekong Delta

Durian is grown in the Mekong Delta and the Central Highlands. More durian is now sold to China, so demand has increased. In the dry season, drought and salt water from the sea can damage durian farms in the Mekong Delta.

The table shows demand and supply in one wholesale durian market.

Price (₫ per kg)	Quantity demanded (tonnes per week)	Quantity supplied (tonnes per week)
40,000	900	100
60,000	700	300
80,000	500	500
100,000	300	700
120,000	100	900

Warm-up **Data skills: use the table before you start.** **no marks**



Plot the demand points (blue) and supply points (red), then join them.

1. Plot the demand and supply curves on the grid.
2. What is the equilibrium price and quantity?
3. At 60,000₫ per kg, is there a shortage or a surplus? How many tonnes?
4. At 120,000₫ per kg, is there a shortage or a surplus? How many tonnes?

State **(a) State what is meant by *equilibrium price*.** **[2]**

To get 2 marks: quantity demanded equals quantity supplied (1) + no shortage or surplus, so no pressure for the price to change (1).

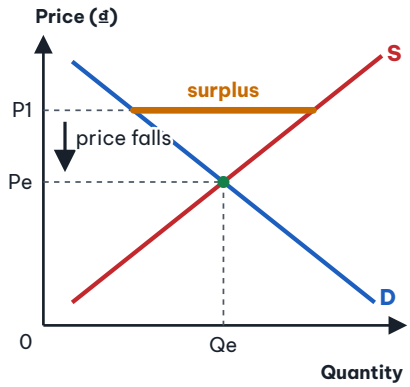
1. **Where:** *Equilibrium price is the price at which _____ equals _____..*
2. **So:** *...so there is no _____ and the price does not change.*

> **Watch out:** say **quantity** demanded and **quantity** supplied. "Where demand equals supply" is weaker.

Explain **(b) Using the table, explain how the market moves from a surplus back to equilibrium.** **[4]**

To get 4 marks: show the surplus with numbers (1) → sellers cut the price (1) → demand extends (1) → supply contracts until Qd = Qs (1).

1. **Surplus with data:** *At 100,000₫ per kg, quantity supplied is _____ tonnes but quantity demanded is only _____ tonnes, a surplus of _____ tonnes.*
2. **Sellers react:** *Sellers have unsold durian, which goes bad quickly, so they _____.*
3. **Buyers react:** *As the price falls, demand _____: buyers want more.*
4. **Sellers react again:** *Supply _____, until Qd = Qs = _____ tonnes at _____₫.*



A surplus pushes the price down to Pe.

Word bank: surplus · unsold stock · cut the price · extends · contracts · equilibrium

> **Watch out:** the curves do **not** shift here. Price changes cause movements **along** the curves.

Analyse **(c) Analyse the effects on the equilibrium price and quantity of durian when demand increases and supply decreases at the same time.** **[6]**

To get 6 marks: both shifts with causes from the context (2) → price definitely rises (2) → quantity uncertain, and why (2). Two diagrams help.

1. **Demand:** *More exports to China increase demand, so the demand curve shifts* _____.
2. **Supply:** *Drought and salt water reduce output, so the supply curve shifts* _____.
3. **Old price:** *At the old price there is now a large* _____.
4. **Price:** *Both changes push the price up, so the equilibrium price* _____ *rises.*
5. **Quantity:** *The effect on quantity is uncertain: the increase in demand pushes quantity* _____, *but the decrease in supply pushes it* _____.
6. **It depends:** *If the demand shift is bigger, quantity* _____. *If the supply shift is bigger, quantity* _____.



Case A: demand shifts more.



Case B: supply shifts more.

> **Watch out:** do not say quantity "stays the same". Say it is **uncertain** and explain what it depends on.

Discuss **(d) Discuss whether the price mechanism is a good way to ration durian when there is a shortage.** **[8]**

To get 8 marks: two developed points for "good" and two for "not good", then a conclusion. Use S.I.R. for the "good" side.

Why it is good

- price rises until the shortage is gone, no queues
- durian goes to people who value it most
- signal and incentive for farmers to grow more

Why it may not be good

- people on low incomes are priced out
- durian trees take years to grow, so incentive is slow
- sellers with market power may hold back stock

1. **Good (point 1):** *When there is a shortage, the price rises until _____, so the shortage disappears without queues or government rules.*
2. **Good (point 2):** *The higher price also acts as a signal and an incentive: _____.*
3. **Not good (point 1):** *However, the price mechanism rations by ability to pay, so _____.*
4. **Not good (point 2):** *Also, durian trees take _____, so in the short run the higher price…*
5. **Conclusion:** *For a non-essential fruit like durian, the price mechanism works quite well, but for essential goods such as _____…*

2.5 SECTION 2.5 Price changes

Key words

Cause of a price change

A change in a condition of demand or a condition of supply shifts a curve. At the old price there is now a shortage or a surplus, so the equilibrium price changes.

Simply: the price changes only when a curve shifts.

nguyên nhân làm giá thay đổi

Increase / decrease in demand

More / less is demanded at every price, because a condition of demand has changed. The demand curve shifts right / left.

Simply: buyers want more (or less) at every price.

cầu tăng (đường cầu dịch chuyển sang phải) · cầu giảm (đường cầu dịch chuyển sang trái)

Increase / decrease in supply

More / less is supplied at every price, because a condition of supply has changed. The supply curve shifts right / left.

Simply: sellers offer more (or less) at every price.

cung tăng (đường cung dịch chuyển sang phải) · cung giảm (đường cung dịch chuyển sang trái)

New equilibrium

The price and quantity where the new demand and supply curves cross, after the price has removed the shortage or surplus.

Simply: where the market settles after the change.

điểm cân bằng mới

Uncertain

When both curves shift, the effect on either price or quantity can be worked out only if we know which shift is bigger. It could rise, fall or stay the same.

Simply: "it depends on which shift is bigger".

không chắc chắn (tùy vào dịch chuyển nào lớn hơn)

Related markets

Markets for substitutes or complements. A change in the price of one product shifts the demand curve for the other product.

Simply: what happens in one market spills into the next one.

thị trường liên quan (hàng hóa thay thế · hàng hóa bổ sung)

Why does the price change only when a curve shifts?

At equilibrium, quantity demanded equals quantity supplied. There is no shortage or surplus, so nothing pushes the price up or down. The price only changes when something **other than the price** shifts a curve and creates a shortage or a surplus at the old price. A movement along a curve is the **result** of a price change, not its cause.

Ở trạng thái cân bằng, không có thiếu hụt hay dư thừa nên giá không đổi. Giá chỉ thay đổi khi một đường dịch chuyển.

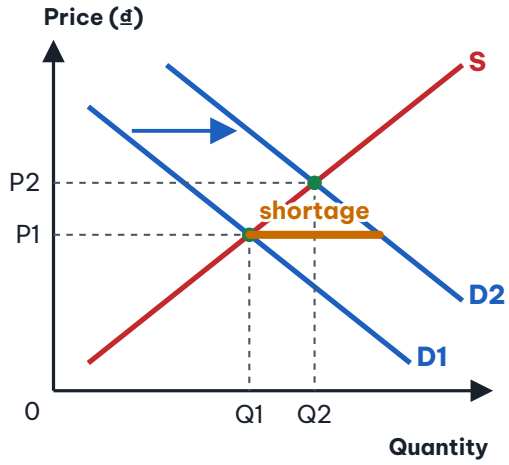
The 5-step method for any price change

1. **What changed?** Name the condition of demand or supply (e.g. income, a substitute's price, costs, weather). Does it affect buyers, sellers, or both?
2. **Which curve, which way?** Demand or supply, right or left.
3. **At the old price:** is there now a shortage or a surplus?
4. **The price moves:** it signals the change, gives an incentive to sellers and rations the product. Demand and supply extend or contract.
5. **New equilibrium:** say what happens to price and quantity (rise, fall or uncertain).

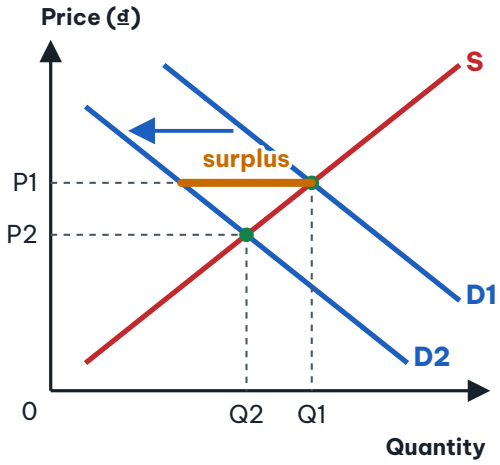
Watch out: most events shift only **one** curve, but some shift both. Always ask whether the event affects buyers, sellers, or both. Example: a heatwave in Hà Nội makes more people want ice cream (demand shifts right), and power cuts mean factories can make less (supply shifts left).

The four basic cases

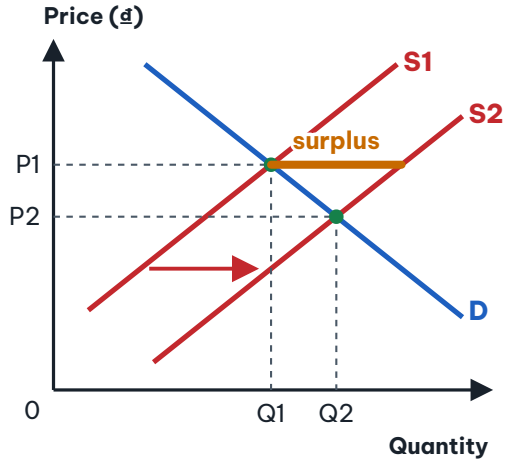
Change	Curve	At the old price	Equilibrium price	Equilibrium quantity
Increase in demand	D shifts right	shortage	rises	rises
Decrease in demand	D shifts left	surplus	falls	falls
Increase in supply	S shifts right	surplus	falls	rises
Decrease in supply	S shifts left	shortage	rises	falls



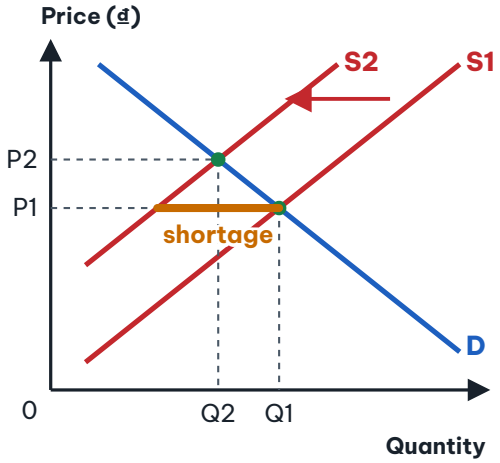
Increase in demand: P and Q rise.



Decrease in demand: P and Q fall.



Increase in supply: P falls, Q rises.



Decrease in supply: P rises, Q falls.

When both curves shift

Think of each shift as a push on price and a push on quantity. If both pushes go the **same** way, the result is certain. If they go **opposite** ways, the result is **uncertain**: it depends on which shift is bigger.

Both changes	Equilibrium price	Equilibrium quantity
Demand increases + supply increases	uncertain	**rises**
Demand decreases + supply decreases	uncertain	**falls**
Demand increases + supply decreases	**rises**	uncertain
Demand decreases + supply increases	**falls**	uncertain



Both increase, demand shifts more: P rises, Q rises.



Both increase, supply shifts more: P falls, Q rises.

Hai đường cùng dịch chuyển: nếu hai lực đẩy cùng hướng thì kết quả chắc chắn; nếu ngược hướng thì không chắc chắn, tùy vào dịch chuyển nào lớn hơn.

Price changes in related markets

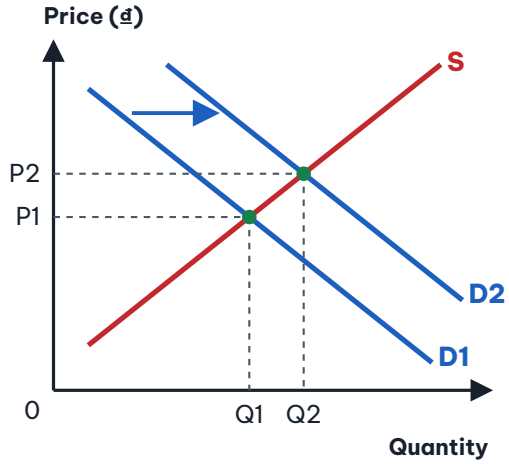
EXAMPLE

Cheaper flights to Phú Quốc

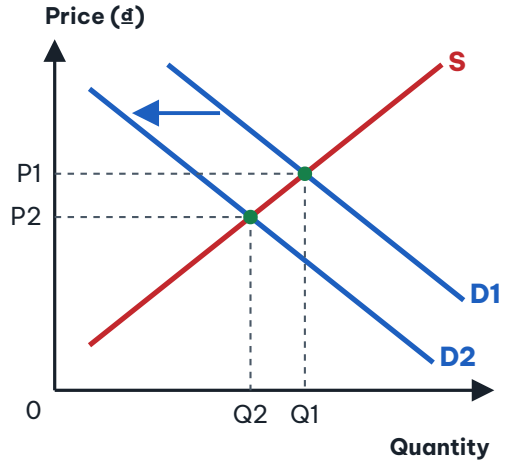
Airlines cut the price of flights from Hồ Chí Minh City to Phú Quốc island. Flights and hotel rooms on the island are **complements**. Flights and the ferry from Rạch Giá are **substitutes**.

Flights cheaper → **more tourists fly** → **demand for hotel rooms shifts right** → **shortage** → **hotel P and Q rise**

Flights cheaper → **some travellers fly instead of taking the ferry** → **demand for ferry tickets shifts left** → **surplus** → **ferry P and Q fall**



Complement: demand for hotel rooms increases.



Substitute: demand for ferry tickets decreases.

Watch out: in the flight market itself, the lower price is a movement **along** the demand curve for flights (an extension). The **shifts** happen in the related markets.

Paper 1 practice: multiple choice

17. Heavy rain destroys many vegetable farms in Đà Lạt. What happens to the equilibrium price and quantity of vegetables?

A Price falls and quantity rises.
B Price rises and quantity rises.
C Price rises and quantity falls.
D Price falls and quantity falls.

18. Demand for a product increases and its supply increases at the same time. What must happen?

A The equilibrium quantity rises.
B The equilibrium price rises.
C The equilibrium price falls.
D The equilibrium quantity falls.

19. The price of smartphones falls. What happens in the market for phone cases?

A Demand shifts left and the price falls.
B Supply shifts right and the price falls.
C Supply shifts left and the price rises.
D Demand shifts right and the price rises.

20. Which change would cause the equilibrium price of bánh mì to fall?

A a rise in the price of phở
B a new bakery opening nearby
C a rise in the cost of flour
D a rise in consumers' incomes

Paper 2 practice: structured question

CONTEXT

Dragon fruit in Bình Thuận

Bình Thuận province grows a lot of dragon fruit (thanh long). Many farmers now light their farms with LED lamps at night. The lamps make the plants flower out of season, so each farm harvests more fruit in a year. Many new farms have also been planted.

At the same time, exports to China have fallen because of stricter quality checks at the border. Watermelon is grown in the same region and sold in the same markets as dragon fruit.

Farm-gate price of dragon fruit (practice figures)	₫ per kg
2024	30,000
2026	12,000

Define **(a) Define *a decrease in demand*.** **[2]**

To get 2 marks: less is demanded at every price (1) + because a condition of demand (not the product's own price) has changed, so the curve shifts left (1).

1. **What:** *A decrease in demand is when _____ is demanded at every _____...*
2. **Why and the curve:** *...because a condition of demand changes, so the demand curve shifts _____.*

> **Watch out:** "people buy less because the price went up" is a **contraction**, not a decrease. Do not put the effects on price in the definition.

Explain **(b) Explain two reasons why the farm-gate price of dragon fruit fell from 30,000₫ to 12,000₫ per kg.** **[4]**

To get 4 marks: for each reason, name it from the context (1) and explain how it shifts a curve and creates a surplus that pushes the price down (1).

1. **Reason 1 (supply):** *LED lamps and new farms mean that more dragon fruit is supplied at every price, so the supply curve shifts* _____.
2. **Explain:** *At the old price of 30,000₫ there is a* _____, *so farmers cut the price.*
3. **Reason 2 (demand):** *Exports to China fall, so less dragon fruit is demanded at every price and the demand curve shifts* _____.
4. **Explain:** *This adds to the* _____, *so the price falls further.*

Word bank: increase in supply | decrease in demand | right | left | surplus | at every price

> **Watch out:** "the price fell because people bought less" is not a reason. Name the condition that changed.

Analyse **(c) Analyse how the fall in the price of dragon fruit could affect the market for watermelon.** **[6]**

To get 6 marks: link the two products (1) → consumers switch (1) → demand for watermelon shifts (1) → surplus at the old price (1) → the price falls and supply contracts (1) → new equilibrium (1).

1. **Link:** *Dragon fruit and watermelon are _____, because…*
2. **Consumers react:** *Dragon fruit is now cheaper, so some consumers _____.*
3. **Curve:** *Demand for watermelon decreases, so the demand curve shifts _____ from D1 to D2.*
4. **Old price:** *At the old price P1 there is a _____ of watermelon.*
5. **Price moves:** *Sellers cut the price. As the price falls, supply _____…*
6. **New equilibrium:** *The new equilibrium price is _____ and the quantity is _____.*



Draw the watermelon market: D1, D2, S, P1, P2, Q1, Q2.

Watch out: the change starts in the dragon fruit market, so in the watermelon market only **demand** shifts. The supply of watermelon does not shift. It contracts along S.

Discuss **(d) Discuss whether the price of dragon fruit will stay low in the future.** **[8]**

To get 8 marks: two developed points for "will stay low" and two for "will not stay low", then a conclusion that says what it depends on.

It might stay low

- LED lamps are a lasting change in technology
- new farms keep producing for many years
- exports to China may stay low if the checks stay strict

It might not stay low

- low profit makes some farmers switch to other crops
- new export markets or new products (juice, dried fruit) raise demand
- a drought or disease cuts supply

1. **Stays low (point 1):** *The price may stay low because LED lamps are new technology that will not go away, so supply stays _____.*
2. **Stays low (point 2):** *Also, exports to China may _____, so demand stays lower.*
3. **Does not stay low (point 1):** *However, at 12,000₫ some farmers earn little profit, so they switch to _____. The number of producers falls, so supply _____.*
4. **Does not stay low (point 2):** *Also, demand could increase if _____.*
5. **Conclusion:** *The price will probably stay below 30,000₫ in the short run, but in the long run it depends on _____.*

BACK PAGES

Answers and mark schemes

Try every question first. Then use these pages to mark your own work. The mark schemes and model answers were written for this guide in the style of IGCSE Paper 2. They are not official Cambridge mark schemes.

Model answer colours: AO1 Knowledge: definitions and facts AO2 Analysis: steps in the chain AO3 Evaluation: weighing up and judging

Paper 1 answer key

Q	Answer	Why
1	**B**	A market brings buyers and sellers together to exchange. Markets do not remove scarcity (D).
2	**D**	Renting a building is a firm hiring a factor of production (land). A, B and C are product markets.
3	**A**	Buyers create demand: they are willing and able to pay. Sellers, not buyers, make goods for profit (C).
4	**C**	Higher demand pushes up the price. The higher price signals to producers to make more.
5	**D**	Only a change in the product's own price causes a movement along its demand curve. A, B and C shift the curve.
6	**A**	Tea is dearer, so people switch to coffee: demand for coffee increases (shifts right).
7	**C**	Own price falls and quantity demanded rises: a movement down along the curve. A is a contraction; B and D are shifts.
8	**B**	Effective demand = wanting a product, backed by the ability to pay.
9	**A**	Better technology lowers costs, so supply increases. B, C and D all decrease supply.
10	**D**	A subsidy lowers costs of production, so supply increases (shifts right).
11	**C**	An extension is a movement along the supply curve, caused only by a change in the product's own price. A, B and D shift the curve.
12	**B**	An indirect tax raises firms' costs, so supply decreases (shifts left).
13	**C**	Equilibrium: quantity demanded = quantity supplied.

14	**A**	Above equilibrium, Qs > Qd: a surplus. Sellers cut the price.
15	**D**	With a shortage the price rises. Demand contracts and supply extends until Qd = Qs. The curves do not shift.
16	**B**	Demand for a normal good increases, so both equilibrium price and quantity rise.
17	**C**	Bad weather decreases supply (shifts left): a shortage at the old price, so price rises and quantity falls.
18	**A**	Both shifts push quantity up, so quantity must rise. They push price in opposite directions, so price is uncertain (B and C may happen, but not for sure).
19	**D**	Phones and cases are complements. More phones are bought, so demand for cases increases and their price rises.
20	**B**	A new bakery is one more firm, so supply increases and the price falls. A and D increase demand; C decreases supply. All three raise the price.

2.1 Electric cars

2.1(a) Define the term market. [2]

Give a mark for	Mark
Buyers and sellers / demanders and suppliers / consumers and producers	**1**
Come together to exchange / trade goods and services (accept: factors of production)	**1**

Accept: "an arrangement", "a system", "online or physical". Do not accept: "a shop" or "a place to buy things" on their own (0).

MODEL ANSWER

A market is any place or system where buyers and sellers come together **[1]** to exchange goods and services. **[1]**

2.1(b) Product market and factor market. [4]

Give a mark for	Mark
Product market: households / consumers buy goods and services from firms	**1**
Example from the context: a family buying an electric car	**1**
Factor market: firms buy or hire factors of production (land, labour, capital) from households / owners	**1**
Example from the context: the car maker hiring engineers or workers, renting land or buying machines	**1**

Do not accept: examples not from the context (max 3), or examples with no explanation of the market.

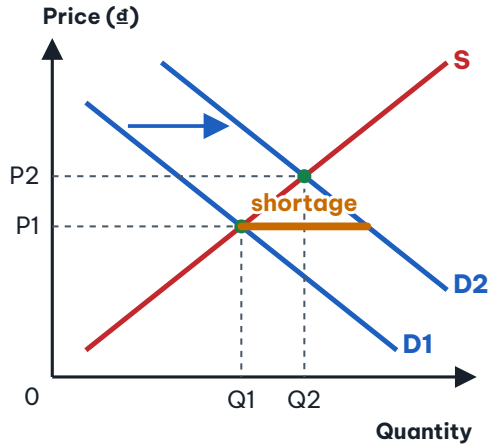
MODEL ANSWER

In a product market, households buy goods and services from firms. **[1]** For example, a family buys an electric car from a car maker. **[1]** In contrast, in a factor market, firms buy or hire factors of production, such as labour and land, from their owners. **[1]** For example, the car maker hires engineers and rents land for a new factory. **[1]**

2.1(c) How the price mechanism allocates resources when demand for electric cars increases. [6]

Give a mark for each link in the chain (max 6)	Mark
Demand increases: the demand curve shifts right (D1 to D2)	**1**
At the old price, Qd > Qs: a shortage (accept: waiting lists, as in the context)	**1**
Signal: the price rises, which tells producers that consumers want more electric cars	**1**
Incentive: higher price / more profit, so firms move more factors of production (workers, machines, land) into making electric cars; supply extends	**1**
Rationing: the higher price makes demand contract, so cars go to those most willing and able to pay	**1**
New equilibrium at a higher price and higher quantity (P2, Q2), with more resources used to make electric cars (and fewer for other products)	**1**

Diagram: a correct, labelled diagram (like the one below) can earn a mark for the shift and a mark for the new P2 and Q2 if they are not written in the answer. Max 6 overall. Do not accept: "supply shifts right" as the response to the higher price. It is an extension.



Demand increases from D1 to D2. Shortage at P1. New equilibrium at P2, Q2.

MODEL ANSWER

When consumer demand for electric cars increases, the demand curve shifts right from D1 to D2. **[1]** At the old price P1, quantity demanded is now greater than quantity supplied, so there is a shortage, which is why there are waiting lists. **[1]** The price rises. This signals to car makers that consumers want more electric cars. **[1]** The higher price gives firms an incentive, because each car earns more profit. So firms hire more engineers, rent more land and buy more machines, and supply extends. **[1]** At the same time, the higher price rations the cars: demand contracts, so the cars go to buyers who are most willing and able to pay. **[1]** A new equilibrium is reached at P2 and Q2, where more scarce resources are used to make electric cars and fewer are used for other products. **[1]**

2.1(d) Will a higher price always lead to more resources being used to make electric cars? [8]

Give marks for	Mark
Why it might: up to 2 points. 1 mark for each point + 1 mark for explaining it. Examples: higher profit is an incentive to hire more factors; the signal attracts new firms into the market; resources move from making petrol cars	**max 4**
Why it might not: up to 2 points. 1 mark for each point + 1 mark for explaining it. Examples: time lag (new factories take years); scarce factors (skilled engineers, batteries) raise costs; firms think the higher demand will not last; government rules or limits	**max 4**

A point with no explanation gets 1 mark, not 2. Always end with a reasoned judgement ("It depends on …"). In the real exam, an 8-mark Discuss answer needs one to reach the top marks.

MODEL ANSWER

A higher price may lead to more resources being used, because each car now earns more profit. This gives car makers an incentive to hire more workers and buy more machines so they can produce more. **[2]** The higher price also signals to other firms that electric cars are profitable. New firms may enter the market, or firms that make petrol cars may move their land, labour and capital into electric cars. **[2]**

However, this may not happen quickly. Building a new car factory can take several years, so in the short run firms cannot increase output much, and the higher price only rations the cars that already exist. **[2]** Also, some resources are very scarce. If there are not enough skilled engineers or batteries, their prices rise, and firms may not be able to expand even though the car price is higher. **[2]** Firms may also wait if they think the higher demand is only temporary.

Overall, a higher price is likely to attract more resources in the long run, but not "always". It depends on how easily factors of production can move and whether firms believe the demand will last.

2.2 Motorbikes

2.2(a) Define effective demand. [2]

Give a mark for	Mark
The wish / willingness / desire to buy a product	1
Backed by the ability to pay / the money to buy it	1

MODEL ANSWER

Effective demand is the willingness to buy a product [1] backed by the ability to pay for it. [1]

2.2(b) Two non-price factors that increase demand for electric motorbikes. [4]

For each of two factors, give	Mark
Names a correct non-price factor	1 + 1
Explains why consumers buy more at every price / why demand shifts right	1 + 1

Accept: rising incomes (normal good); a rise in the price of petrol (petrol is a complement to petrol motorbikes, so people switch to the substitute); a rise in the price of petrol motorbikes (substitute); a fall in the cost of charging (complement); tastes and fashion (modern, clean); advertising; population growth; lower interest rates / cheaper loans; city limits on petrol motorbikes. Do not accept: a fall in the price of electric motorbikes (that is an extension, not an increase).

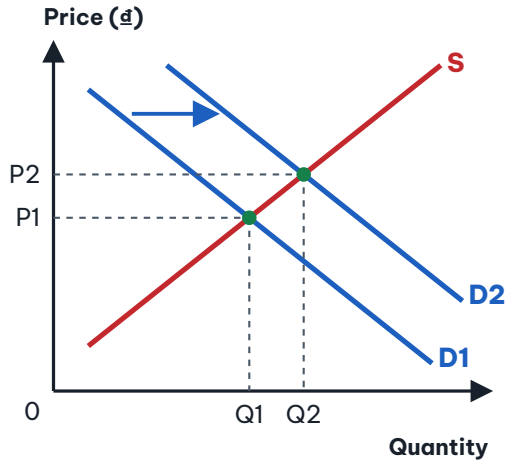
MODEL ANSWER

One factor is rising incomes. [1] Electric motorbikes are a normal good, so when people earn more they can afford to buy more of them at every price, and the demand curve shifts to the right. [1] A second factor is a change in tastes. [1] Young people see electric motorbikes as modern and clean, so more of them want to buy one at each price, and demand increases. [1]

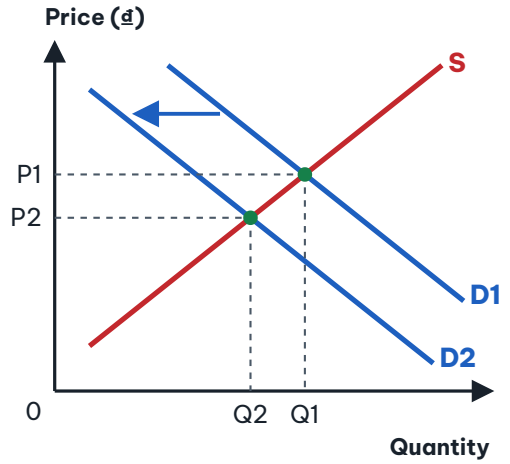
2.2(c) A fall in the price of petrol: petrol and electric motorbike markets. [6]

Give a mark for each link (max 6)	Mark
Petrol and petrol motorbikes are complements, and/or petrol motorbikes and electric motorbikes are substitutes	**1**
Cheaper petrol makes petrol motorbikes cheaper to run	**1**
Demand for petrol motorbikes increases / shifts right	**1**
Equilibrium price and quantity of petrol motorbikes rise	**1**
Consumers switch away from electric motorbikes: demand decreases / shifts left	**1**
Equilibrium price and quantity of electric motorbikes fall	**1**

Do not accept: "supply of petrol motorbikes increases". The change is in demand. Correct labelled diagrams can earn the price and quantity marks.



Petrol motorbikes: demand increases, P and Q rise.



Electric motorbikes: demand decreases, P and Q fall.

MODEL ANSWER

Petrol and petrol motorbikes are complements, because they are used together. Petrol motorbikes and electric motorbikes are substitutes. **[1]** A fall in the price of petrol makes petrol motorbikes cheaper to run. **[1]** So demand for petrol motorbikes increases, and the demand curve shifts right from D1 to D2. **[1]** This creates a shortage at the old price, so the equilibrium price of petrol motorbikes rises from P1 to P2 and the quantity rises from Q1 to Q2. **[1]** Because petrol motorbikes are now cheaper to use, some consumers switch away from electric motorbikes, so demand for electric motorbikes decreases and the curve shifts left. **[1]** This creates a surplus at the old price, so the equilibrium price and quantity of electric motorbikes both fall. **[1]**

2.2(d) Will rising incomes increase demand for motorbikes? [8]

Give marks for	Mark
Why it might: up to 2 points, each 1 + 1. Examples: motorbikes are a normal good for most families; more young people can afford their own bike; people buy newer or electric models	**max 4**
Why it might not: up to 2 points, each 1 + 1. Examples: richer families switch to cars (the motorbike acts as an inferior good for them); substitutes such as the metro or ride-hailing cars; city limits on petrol motorbikes; traffic jams; income rises may go mainly to people who already own motorbikes	**max 4**

MODEL ANSWER

Rising incomes may increase demand for motorbikes because, for most families in Việt Nam, a motorbike is a normal good. With more income, people can afford to buy a motorbike at each price, so the demand curve shifts right. **[2]** Higher incomes also mean that more young people, such as students who start work, can afford their own motorbike instead of sharing one with their family. **[2]**

However, demand may not increase for everyone. As incomes rise further, some families may buy a car instead, because a car is safer and more comfortable in the rain. For these families, the motorbike acts as an inferior good, so their demand for motorbikes falls. **[2]** Also, other conditions of demand may change at the same time. The new metro in Hồ Chí Minh City is a substitute, and some cities plan to limit petrol motorbikes, so demand could fall even if incomes rise. **[2]**

Overall, demand for motorbikes will probably rise for lower- and middle-income families but may fall for richer families. The total effect depends on how high incomes rise and how good the substitutes are.

2.3 Coffee and laptops

2.3(a) Define the term supply. [2]

Give a mark for	Mark
The quantity / amount producers are willing and able to sell	1
At each / a given price, in a given time period	1

MODEL ANSWER

Supply is the quantity of a product that producers are willing and able to sell [1] at each price in a given time period. [1]

2.3(b) Two factors that decrease the supply of coffee beans. [4]

For each of two factors, give	Mark
Names a correct condition of supply	**1 + 1**
Explains why producers supply less at every price / supply shifts left	**1 + 1**

Accept: drought / bad weather; higher fertiliser prices or wages; fewer farmers (switching to durian); a new indirect tax; removal of a subsidy; pests or disease; lower productivity. Do not accept: a fall in the price of coffee (that is a contraction of supply).



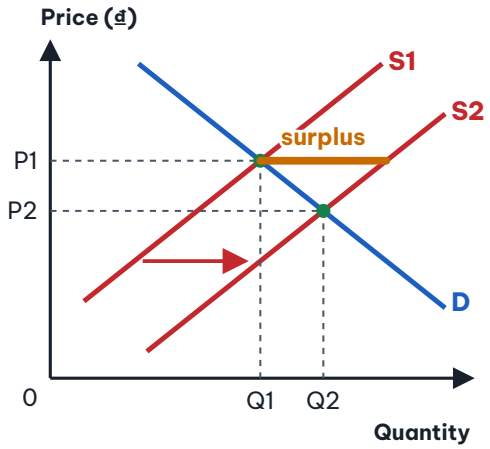
Supply decreases from S1 to S2: price rises, quantity falls.

MODEL ANSWER

One factor is bad weather, such as a drought in the Central Highlands. **[1]** The drought damages coffee trees, so farmers can harvest fewer beans at every price, and the supply curve shifts left. **[1]** A second factor is a rise in costs of production: fertiliser has risen from 600,000₫ to 750,000₫ a bag and wages have also risen. **[1]** Each kilogram of coffee now costs more to grow, so it is less profitable, and farmers supply less at each price. **[1]**

2.3(c) Robots in factories: supply and price of laptops. [6]

Give a mark for each link (max 6)	Mark
Robots are new technology / raise productivity	1
Cost of producing each laptop falls	1
Firms supply more at every price: supply shifts right (S1 to S2)	1
At the old price, Qs > Qd: a surplus	1
Firms cut prices; as the price falls, demand extends (and supply contracts along S2)	1
New equilibrium: lower price (P2) and higher quantity (Q2)	1



Supply increases from S1 to S2. Surplus at P1. New equilibrium at P2, Q2.

MODEL ANSWER

Robots are a new technology, so each worker can produce more laptops per hour: productivity rises. **[1]** This lowers the cost of producing each laptop. **[1]** With lower costs, firms earn more profit at every price, so they are willing and able to supply more, and the supply curve shifts right from S1 to S2. **[1]** At the old price P1, quantity supplied is now greater than quantity demanded, so there is a surplus. **[1]** Firms cut their prices to sell their stock, and as the price falls, demand extends. **[1]** A new equilibrium is reached at a lower price P2 and a higher quantity Q2. **[1]**

2.3(d) Will a subsidy to coffee farmers increase the supply of coffee? [8]

Give marks for	Mark
Why it might: up to 2 points, each 1 + 1. Examples: lower costs → more profit → supply shifts right; money for better plants, fertiliser or irrigation raises productivity; keeps farmers in coffee instead of durian	**max 4**
Why it might not: up to 2 points, each 1 + 1. Examples: coffee trees take 3 to 4 years to produce beans; drought may cut supply more than the subsidy raises it; durian may still be more profitable; the subsidy may be too small or not reach small farmers; farmers may spend it on other things	**max 4**

MODEL ANSWER

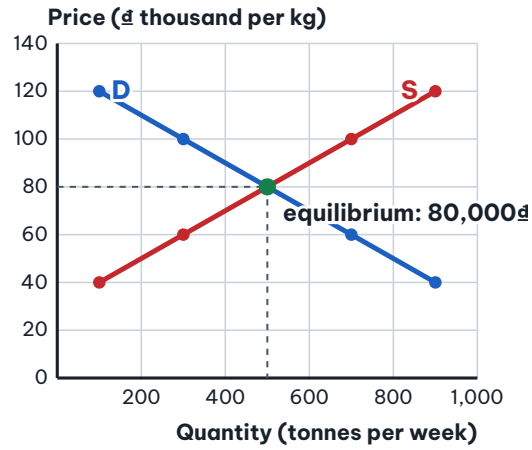
A subsidy may increase supply because it lowers farmers' costs of production. Each kilogram of coffee becomes more profitable, so farmers are willing to supply more at every price, and the supply curve shifts right. **[2]** Farmers could also use the money to buy better plants, fertiliser or water pumps. This raises productivity, so they can grow more coffee on the same land. **[2]**

However, the effect may be slow. New coffee trees take three to four years to produce beans, so in the short run supply may hardly change. **[2]** Also, other factors may cancel out the subsidy. A serious drought could cut the harvest by more than the subsidy adds, and if durian is still more profitable, some farmers will keep switching away from coffee. **[2]**

Overall, a subsidy will probably increase supply in the long run if it is large enough and reaches small farmers, but weather and the time trees take to grow mean the short-run effect may be small.

2.4 Durian

Data skills warm-up



The plotted durian market.

1. See the graph.
2. Equilibrium: **80,000₫ per kg** and **500 tonnes** per week.
3. At 60,000₫: Qd = 700, Qs = 300. **Shortage of 400 tonnes.**
4. At 120,000₫: Qd = 100, Qs = 900. **Surplus of 800 tonnes.**

2.4(a) State what is meant by equilibrium price. [2]

Give a mark for	Mark
The price where quantity demanded = quantity supplied (accept: where the curves cross)	**1**
No shortage or surplus / the market clears / no pressure for the price to change	**1**

MODEL ANSWER

Equilibrium price is the price at which quantity demanded equals quantity supplied, **[1]** so there is no shortage or surplus and no pressure for the price to change. **[1]**

2.4(b) From a surplus back to equilibrium. [4]

Give a mark for	Mark
Uses the table to show a surplus: e.g. at 100,000₫, Qs 700 > Qd 300, surplus 400 tonnes (accept 120,000₫: surplus 800)	**1**
Sellers have unsold stock, so they cut the price	**1**
As the price falls, demand extends (Qd rises)	**1**
Supply contracts (Qs falls) until Qd = Qs at 80,000₫ and 500 tonnes	**1**

Do not accept: "demand shifts right" or "supply shifts left". Nothing shifts here.

MODEL ANSWER

At 100,000₫ per kg, sellers want to sell 700 tonnes but buyers only want 300 tonnes, so there is a surplus of 400 tonnes. **[1]** Sellers have unsold durian, which goes bad quickly, so they cut the price. **[1]** As the price falls, demand extends because more buyers can afford durian. **[1]** At the same time supply contracts because a lower price is less profitable. This continues until quantity demanded equals quantity supplied at 80,000₫ and 500 tonnes. **[1]**

2.4(c) Demand increases and supply decreases at the same time. [6]

Give a mark for	Mark
Demand increases / shifts right, with a reason (exports to China)	1
Supply decreases / shifts left, with a reason (drought, salt water)	1
A shortage at the old price, so the price is pushed up	1
The equilibrium price definitely rises (both changes push it up)	1
The effect on quantity is uncertain: demand pushes it up, supply pushes it down	1
Quantity rises if the demand shift is bigger; falls if the supply shift is bigger (accept diagrams showing this)	1



Case A: demand shifts more. P rises, Q rises.



Case B: supply shifts more. P rises, Q falls.

MODEL ANSWER

More durian is sold to China, so demand increases and the demand curve shifts right from D1 to D2. **[1]** At the same time, drought and salt water damage farms, so supply decreases and the supply curve shifts left from S1 to S2. **[1]** At the old price, quantity demanded is now much greater than quantity supplied, so there is a shortage and buyers push the price up. **[1]** Because both changes push the price up, the equilibrium price definitely rises from P1 to P2. **[1]** The effect on quantity is uncertain. The increase in demand pushes quantity up, but the decrease in supply pushes it down. **[1]** If the increase in demand is bigger than the decrease in supply, quantity rises (Case A). If the decrease in supply is bigger, quantity falls (Case B). **[1]**

2.4(d) Is the price mechanism a good way to ration durian in a shortage? [8]

Give marks for	Mark
Why it is good: up to 2 points, each 1 + 1. Examples: the price rises until Qd = Qs, so the shortage ends without queues or rules; durian goes to those who value it most (most willing and able to pay); the higher price signals to farmers and gives an incentive to grow more, so the shortage gets smaller over time	**max 4**
Why it may not be good: up to 2 points, each 1 + 1. Examples: rationing is by ability to pay, so people on low incomes are priced out; this is unfair for essential goods; durian trees take years to grow, so the incentive works slowly; sellers or traders may hold back stock to push the price higher	**max 4**

MODEL ANSWER

The price mechanism can be a good way to ration durian. When there is a shortage, the price rises until quantity demanded equals quantity supplied, so the shortage disappears without long queues or government rules. **[2]** The higher price also signals to farmers that durian is scarce, and the extra profit gives them an incentive to plant more trees or bring durian from other regions, so the shortage gets smaller over time. **[2]**

However, the price mechanism rations by ability to pay, not by need. Families on low incomes may want durian but can no longer afford it, so they are priced out of the market. **[2]** Also, durian trees take several years to start producing fruit, so in the short run the higher price cannot increase supply much. It mainly moves money from buyers to sellers. **[2]**

Overall, for a non-essential fruit like durian, the price mechanism works quite well, because nobody is harmed by going without durian. For essential goods such as rice or medicine, rationing by price is less fair, and the government may need to step in.

2.5 Dragon fruit

2.5(a) Define a decrease in demand. [2]

Give a mark for	Mark
Less is demanded at every / each price (accept: consumers are willing and able to buy less at each price)	**1**
Caused by a change in a condition of demand / a non-price factor, so the demand curve shifts left	**1**

Do not accept: "people buy less because the price rises" (that is a contraction, 0). Effects such as "so the price falls" earn no marks.

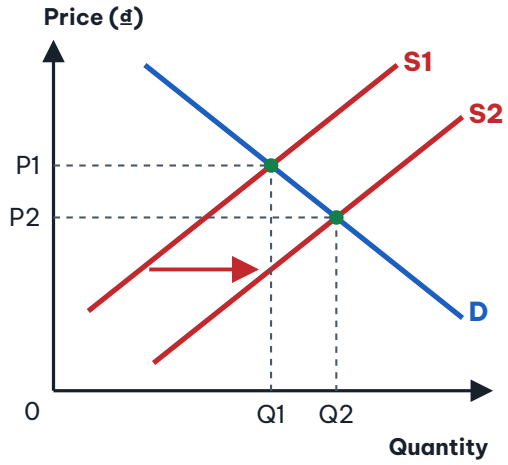
MODEL ANSWER

A decrease in demand is when less of a product is demanded at every price, **[1]** because a condition of demand (not the product's own price) has changed, so the demand curve shifts left. **[1]**

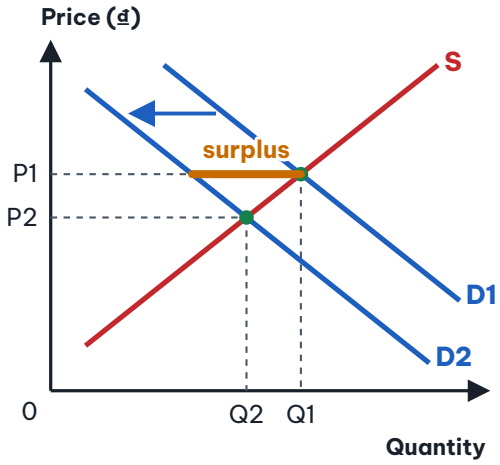
2.5(b) Two reasons why the price of dragon fruit fell. [4]

For each of two reasons, give	Mark
Names a reason from the context: an increase in supply (LED lamps / out-of-season flowering / new farms) or a decrease in demand (fewer exports to China / stricter border checks)	**1 + 1**
Explains it: the curve shifts (supply right / demand left), so there is a surplus at the old price and sellers cut the price	**1 + 1**

Accept: LED lamps and new farms as two separate supply reasons, if each is explained. Do not accept: "people bought less" or "farmers sold more" with no cause (these describe movements, not reasons).



Reason 1: supply increases from S1 to S2.



Reason 2: demand decreases from D1 to D2. Surplus at P1.

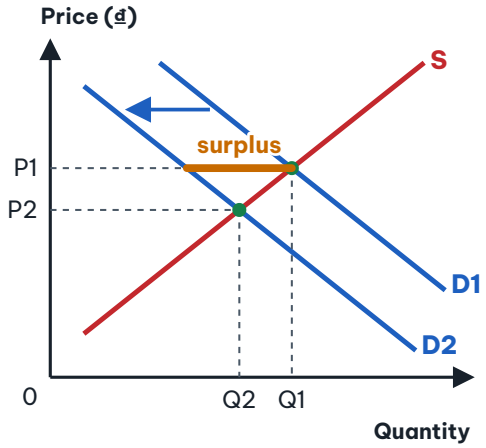
MODEL ANSWER

One reason is an increase in supply. LED lamps make the plants flower out of season and many new farms have been planted. **[1]** So more dragon fruit is supplied at every price and the supply curve shifts right. At the old price of 30,000₫ there is a surplus, so farmers cut the price to sell their fruit. **[1]** A second reason is a decrease in demand, because exports to China have fallen after stricter border checks. **[1]** Less dragon fruit is demanded at every price, so the demand curve shifts left. This makes the surplus even bigger and pushes the price down further, to 12,000₫. **[1]**

2.5(c) The effect on the market for watermelon. [6]

Give a mark for each link in the chain (max 6)	Mark
Dragon fruit and watermelon are substitutes (both are fruit sold in the same markets)	1
Dragon fruit is cheaper, so some consumers buy dragon fruit instead of watermelon	1
Demand for watermelon decreases: the demand curve shifts left (D1 to D2)	1
At the old price, Qs > Qd: a surplus of watermelon	1
Sellers cut the price; supply contracts (and demand extends along D2)	1
New equilibrium at a lower price (P2) and a lower quantity (Q2)	1

Diagram: a correct, labelled diagram can earn the shift mark and the new equilibrium mark if they are not written. Do not accept: "supply of watermelon shifts left" as the main effect.



Demand for watermelon decreases from D1 to D2. Surplus at P1. New equilibrium at P2, Q2.

MODEL ANSWER

Dragon fruit and watermelon are substitutes, because consumers can buy one fruit instead of the other. **[1]** Dragon fruit is now much cheaper, so some consumers switch from watermelon to dragon fruit. **[1]** This means that less watermelon is demanded at every price, and the demand curve for watermelon shifts left from D1 to D2. **[1]** At the old price P1, quantity supplied is now greater than quantity demanded, so there is a surplus of watermelon. **[1]** Sellers cut the price to sell their stock. As the price falls, supply contracts because watermelon is less profitable. **[1]** As a result, a new equilibrium is reached at a lower price P2 and a lower quantity Q2. **[1]**

2.5(d) Will the price of dragon fruit stay low? [8]

Give marks for	Mark
Why it might stay low: up to 2 points, each 1 + 1. Examples: LED lamps are a lasting change in technology, so supply stays higher; new plants keep fruiting for many years, so the number of farms will not fall quickly; demand from China may stay low if checks stay strict or China grows more of its own fruit	**max 4**
Why it might not stay low: up to 2 points, each 1 + 1. Examples: low profit makes some farmers switch to other crops (such as durian), so the number of producers falls and supply decreases; new export markets, better quality or new products (juice, dried fruit) increase demand; a drought or plant disease cuts supply	**max 4**

A point with no explanation gets 1 mark, not 2. A point must say which curve shifts and why. Always end with a reasoned judgement ("It depends on …"). In the real exam, an 8-mark Discuss answer needs one to reach the top marks.

MODEL ANSWER

The price may stay low because LED lamps are a new technology that farmers will keep using. Each farm will keep producing more fruit every year, so supply will stay high and the price will stay near its new, lower equilibrium. **[2]** Demand may also stay low. If the border checks stay strict, exports to China will not recover, so the demand curve will stay further to the left. **[2]**

However, the low price may not last. At 12,000₫ per kg many farmers earn very little profit, so some will switch their land to more profitable crops such as durian. The number of producers falls, so supply decreases and the price rises again. **[2]** Also, demand could increase. If farmers meet the quality checks, or sell to new markets or make products such as dragon fruit juice, more is demanded at every price, and the price rises. **[2]**

Overall, the price will probably stay below 30,000₫ in the short run, because the extra supply from lamps and new farms will not disappear quickly. In the long run it depends on how many farmers leave and whether new markets are found.

IGCSE Economics study companion, sections 2.1 to 2.5 (school numbering: 2.3 Supply, 2.4 Price determination, 2.5 Price changes). Practice figures in the contexts are for exam practice only. Mark schemes are written for this guide and are not official Cambridge material.